

HUSSAIN HABIB

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EDUCATION

Lahore University of Management Sciences (LUMS)

2023 - 2027

B.S. in Computer Science

CGPA: 3.790

Minor in Robotics

Relevant Coursework: Calculus I & II, Machine Learning, Computer Vision, Junior Design Studio (Robotics), Multi-Agent Systems, Mobile Robotics, Deep Learning

PUBLICATIONS

UAV-Based Crop Reconstruction and Trait Estimation Using RGB Imagery Without External Geospatial Infrastructure

Muhammad Hussain Habib Chaudhry, Muhammad Ibrahim Rana, and Hassan Jaleel

Computers and Electronics in Agriculture, Elsevier, 2026.

DOI: [10.1016/j.compag.2026.111677](https://doi.org/10.1016/j.compag.2026.111677)

PROFESSIONAL EXPERIENCE

Lahore University of Management Sciences

July 2025 - Present

Research Assistant - Robotics @ NARL

Supervisor: Dr. Hassan Jaleel

- Developed end-to-end photogrammetry pipelines covering autonomous flight data collection, frame extraction, image alignment, dense point cloud generation, mesh reconstruction, and orthomosaic creation
- Engineered Digital Elevation Maps (DEMs) for canopy height estimation and multi-stage crop health assessment including leaf discoloration detection and panicle counting using open-source computer vision models (SAM, SegVeg, YOLOv5)
- Conducted crop yield prediction research using automatic feature extraction from UAV RGB imagery and Digital Elevation Models (DEMs), with spectral features demonstrating strong correlation with ground-truth yield measurements

Lahore University of Management Sciences

June - July 2025 & June - July 2026

Teaching Assistant

Instructor: Dr. Malik Jahan

- Served as Teaching Assistant for "Computational Problem Solving (CS-100)" course for approximately 80 students
- Assisted students in understanding programming concepts during lab sessions and office hours
- Created and evaluated quizzes and lab assignments, and invigilated examinations

Lahore University of Management Sciences

May 2024 - September 2024

Research Intern

Supervisor: Dr. Hassan Jaleel

- Developed and implemented advanced algorithms using ROS (Robot Operating System) for simulated environments in Gazebo and real-world robotic platforms
- Gained experience in autonomous navigation, sensor integration, and algorithm optimization

TECHNICAL SKILLS

Programming Languages	Python, C++, Java, JavaScript/TypeScript
Robotics & CV	ROS, ROS2, OpenCV, YOLO, Nav2, SLAM Toolbox
Web & Visualization	React, Three.js, Vite, Node.js
Tools & Platforms	Gazebo Harmonic, Git, Linux, SolidWorks, MATLAB, Proteus, Firebase

OPEN-SOURCE PROJECTS

BAGEL: ROS Bag File Web Visualizer 2026 - Present

github.com/Hussain004/BAGEL · bagel-ros2.vercel.app

- Built and shipped an open-source, fully client-side ROS1/ROS2 bag file visualizer supporting .mcap, .db3, .bag, point clouds, and live robot data streaming, with no installation required
- Directed feature scope and architecture across 19 releases, including 3D scene rendering and URDF overlays

Forma: Browser-Native ONNX/TFLite Model Visualizer & Editor 2026 - Present

github.com/Hussain004/Forma · forma-ml.vercel.app

- Built an open-source, browser-native visualizer and editor for ONNX/TFLite neural network graphs, supporting structural editing and export with no Python or server dependency
- Directed design of the graph inspector, quantization estimator, and inference benchmarking

ACADEMIC PROJECTS

Open-Vocabulary Semantic Room Navigation Spring 2026

EE3002 Junior Design Studio, LUMS

- Designed and deployed a distributed architecture offloading YOLO-World inference to a laptop base station while running Nav2 and hardware drivers on a Raspberry Pi 5
- Implemented an early semantic exit strategy using LiDAR and RGB input to evaluate room scores dynamically, avoiding exhaustive search and mapping contexts to a persistent spatial memory

3D Scene Reconstruction and Virtual Tour Fall 2025

CS436 Computer Vision, LUMS

- Built a full Structure-from-Motion pipeline from scratch (SIFT feature extraction, Essential Matrix estimation, incremental PnP pose registration, triangulation, and sparse bundle adjustment) producing a ~200,000-point 3D reconstruction across 30 frames
- Delivered an interactive web-based virtual tour in Three.js with teleportation, waypoint navigation, and photograph overlays aligned to reconstructed camera poses

AWARDS & ACHIEVEMENTS

- Placed on Dean's Honor List, LUMS - Junior Year
- Placed on Dean's Honor List, LUMS - Sophomore Year
- Placed on Dean's Honor List, LUMS - Freshman Year
- Full Merit Scholarship for A Levels in International School Lahore (ISL)